$W = p\Delta V = rpV$

20 C

$$E = \frac{Q}{4\pi\varepsilon_0 r^2} = 32.0 \quad (r\text{: side of 1 square})$$

$$E_s = \frac{Q}{4\pi\varepsilon_0 (3r)^2} = \frac{1}{9}(32.0) = 3.5556 = 3.6 \text{ N C}^{-1}$$

$$E' = \frac{Q}{4\pi\varepsilon_0 (\sqrt{2}r)^2} = \frac{1}{2}E = 16.0 \text{ N C}^{-1}$$

$$E_R = \frac{Q}{4\pi\varepsilon_0 (\sqrt{4r^2 + r^2})^2} = \frac{Q}{4\pi\varepsilon_0 (\sqrt{5}\ r)^2}$$

$$= \frac{1}{5}E = 6.4 \text{ N C}^{-1}$$

21 D Since the sphere is a conductor, the charge will reside only on the outermost surface of the